

Customer Information

Freeheart Properties

+233200000000

freeheart.abokobi@example.com

Freeheart, Abokobi, Greater Accra

Abokobi, Ghana

Quote #: QT-FREEHEART-GLASS-91D54E

Date: May 15, 2026

Valid for: 30 days

Hybrid Solar Package

Hybrid solar for ~14 kWh/day at Abokobi (Accra sun data). 8× 570W panels, 5.0 kW hybrid inverter, 5.0 kWh storage (8h essential backup).

Item	Unit price (GHS)	Qty	Amount (GHS)
Longi 570W Panel	1,500.00	8.0	12,000.00
PV mid / end clamp (each) (3.5× panels)	30.00	28.0	840.00
MC4 connector pair (~1.2 pairs per panel)	25.00	10.0	250.00
Energy Precision 5kW 5.0kW Inverter	6,500.00	1.0	6,500.00
RR 10MM Copper Cable (100m coil) (AC feeder coils vs inverter kW)	2,500.00	2.0	5,000.00
AC isolator switch 40A (AC isolator per inverter)	280.00	1.0	280.00
100A changeover (1 per inverter / genset path)	1,500.00	1.0	1,500.00
LiFePO4 5.0kWh Battery	13,000.00	1.0	13,000.00
Battery cable 35mm copper (per meter) (~2 m per kWh)	80.00	10.0	800.00
250A MCCB battery breaker (battery bank protection)	650.00	2.0	1,300.00
Mounting structure (rails) (proforma rail sets)	350.00	5.0	1,750.00
Transportation of logistics	3,000.00	1.0	3,000.00
Installation & commissioning (standard) (tier for 4.6 kW)	6,000.00	1.0	6,000.00
Single core 4mm PV cable (100m coil) (PV homerun (1 DC strings))	2,500.00	2.0	5,000.00
AC cable 6 mm ² (100m coil) (AC branch / DB wiring)	1,800.00	1.0	1,800.00
63A DC MCB 2P (DC string groups (1 strings))	200.00	2.0	400.00

Item	Unit price (GHS)	Qty	Amount (GHS)
DC isolator switch 32A (DC isolator per string group)	350.00	1.0	350.00
100A AC breaker (AC branch protection)	250.00	1.0	250.00
6-way breaker box	180.00	1.0	180.00
13-way breaker box	350.00	2.0	700.00
SPD / lightning arrester (500V class) (standard hybrid)	300.00	1.0	300.00
Earthing kit (rod, clamp, 16mm earth cable) (earthing per array section)	450.00	2.0	900.00
PVC trunking 25mm (per metre) (trunking metres)	35.00	12.0	420.00
DC/AC labeling & warning kit (labels & warnings)	120.00	1.0	120.00
Miscellaneous (MC4, trunking, PVC, glands, fixings, lugs, ties) (remaining consumables (glands, lugs, ties))	1,500.00	1.0	1,500.00

Sub total: **64,140.00**

Tax (12.5%): **8,017.50**

Discount (0%): **0.00**

Option total (GHS): **72,157.50**

SYSTEM SPECIFICATIONS & ENGINEERING DETAILS

DAILY ENERGY REQUIREMENT

13.55 kWh/day

EFFECTIVE DAILY ENERGY

18.82 kWh/day

After system losses

MONTHLY GENERATION

564.6 kWh

PEAK SUN HOURS

5.0 hrs/day

Location: N/A

PV ARRAY CAPACITY

4.56 kW

8 x 570W panels

PANEL BRAND

Longi

570W model

INVERTER CAPACITY

3.5 kW

Selected: 5.0kW
(for quote)

DC/AC RATIO

0.91

Optimized

BATTERY CAPACITY

5.0 kWh

BACKUP DURATION

8 hours

REQUIRED ROOF AREA

25.0 m²

MOUNTING RAILS (EST.)

5 sticks (18 ft)

≈ 22.6 m linear

SYSTEM EFFICIENCY

72.0 %

Engineering Notes: This system is designed using industry-standard engineering factors, including system efficiency (accounting for inverter, wiring, and temperature), an optimized DC/AC ratio to prevent inverter clipping, and location-specific peak sun hours for accurate energy-generation calculations. All calculations include appropriate safety margins for reliable performance.

Mounting rail stick count is a **planning estimate** (18 ft segments, portrait grid); confirm with site layout and structural design.

Terms & Conditions

This quotation is valid for 30 days from the date of issue. All prices are subject to change without notice. Installation and project delivery terms will be discussed upon acceptance.

Payment Terms

30% deposit · 40% on delivery · 30% on commissioning

Notes

Site: Freeheart, Abokobi — glass building only (3 floors). Down floor: 8 lights, 4 sockets, 2 fans. 1st floor: 7 lights, 4 sockets, 2 fans. 2nd floor: 9 lights, 7 sockets, 2 fans. Assumptions: 12W LED, 150W diversified outlets, 70W fans. Contact: Site Contact — Abokobi +233200000000

Thank you for your business!